# Deep Learning - COSC2779/2972

## Introduction to Deep Learning

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# Outline

1. Course Overview
2. Introduction to Deep Learning
3. Review of Machine Learning Fundamentals

# Teaching Team



**A/Prof. Tennakoon**

Course Coordinator



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Tutor / Lab Demo



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# Objectives of Our Course

Learn about Deep learning (DL) conceptually and practically.

- Different problem types of DL and sample of approaches.
- Learn about the typical process of developing deep Neural Networks.
- Apply these to solving real world prediction and exploratory tasks.

Focus on:

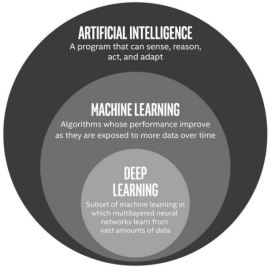
- Moving beyond "black box" AI to build models we can verify and explain.
  - *Analysis* of what the algorithms are doing.
  - *Analysis* of the data set and results.
- Learning an initial set of tools, the *key DL techniques*.
- *Terminology and concepts* so can learn other DL approaches not covered in course.

Official Prerequisite:

- (Computational) Machine Learning - COSC2793/2673

Desirable:

- Knowledge of Python
- Calculus (gradients)
- Linear Algebra
- Optimisation
- Probability and Statistics
- Algorithms




Machine Learning: Statistics, Algorithm & complexity theory, Optimization.

**See Canvas Week 1 Optional readings for some revision materials.**

# Course Structure

| | | |
|---|---|---|
| **Week 1** | Introduction Deep Learning | |
| **Week 2** | Deep Feed Forward Networks | Basics |
| **Week 3** | Neural Network Optimization | |
| **Week 4** | Convolutional Neural Networks | |
| **Week 5** | Vision Application & CNN Architectures | Vision |
| **Week 6** | Practical methodology/Hyper parameter tuning | |
| **Week 7** | Modeling Sequential Data | |
| **Week 8** | Time Series Applications | Sequence pred. |
| **Week 9** | Unsupervised Learning/Generative Models | |
| **Week 10** | Representation Learning/Self Supervised Learning | |
| **Week 11** | Neural Network Model Interpretation/Explainable AI (XAI) | Advanced Topics |
| **Week 12** | Review | |

- **Delivery mode:** On-campus
- **Classroom and Hours:**
  - Pre-Recorded Content: Recording will get you ready for the lectorial.
  - Lectures (Lectorial): Will also contain information that is not covered in the pre-recorded videos.
  - Lab/Tute: On-Campus
- **Canvas:** Everything (slides/homeworks/projects/discussion board etc.)

# Delivery Plan

- **Lectures:** On-Campus
  - Each week a set of pre-recorded videos (weeks 1-8 only) or reading materials will be a available prior to the lecture. Please go through them before joining the lecture.
  - There will be a Lecture session during the timetable slot.
  - Lecture will cover summary of pre-recorded videos, **additional material**, examples and Q&A.
- **Lab/Tutorials:** On-Campus
  - Each week lab exercises and tutorial questions will be made available prior to class. Please go through them before joining and be prepared to discuss the ideas.
  - *Labs will be a good place to ask questions on assignments.*
- **Self Study** You are also expected to spend a significant amount of time in private study, working through the course as presented in classes and learning materials and gaining practice at solving conceptual and technical problems.
- **Office hours (consultation):** See canvas for details.

# Course Workload and Assignments

**Workload (every week):** $>$ 10**hrs:**

- Four Contact hours: Lecture 2hrs, Lab/Tutorial 2hr
- At least six Self study hours (include pre recorded videos)

**Assignments:**

- DL Project - computer vision (30%) - Due Friday Week 6
- DL Project - sequence modeling (30%) - Due Friday Week 11
- In Campus Exam (40%) Will be scheduled in week 14.

**Late penalty for assignments**

- Standard CS, -10% each day.
- Set last day for submission.
- Special Consideration may result in equivalent assessment, **not extensions**.

**Getting Help Before Submission:**

- Use consultation hours (lecturers/tutors) and lab times to get feedback on your project progress. While we won't give you the direct answers, we are always happy to point you in the right direction!

**Feedback and Grades:**

- We aim to provide assignment feedback in a timely manner (typically 2–3 weeks from the due date). A formal mechanism will be provided if you need to request a review of your grades.

**AI-Assisted Feedback Generation:**

- This course uses an AI agent to assist markers in preparing detailed feedback reports.
- **Important:** Human markers are still entirely responsible for evaluating and grading your work. The AI is strictly used to improve the quality, detail, and standardization of the written feedback you receive.

# Text and Reference Books

Primary reference:

- Ian Goodfellow, Yoshua Bengio, Aaron Courville, *Deep Learning*, MIT Press, 2016, http://www.deeplearningbook.org.

Other references:

- Tom Mitchell, *Machine Learning*, McGraw Hill, 1997.
- Neural Networks and Deep Learning by Michael Nielsen Link

Additional Reading material (references) specific to each week will be made available via the relevant canvas module.

# Expectations

**Us $\rightarrow$ You**

- This is an elective (for most of you). Hence we do expect you to work hard and be highly motivated.
- Please attend classes.
- Interact while in class! Be curious, ask questions.
- We try to be fair as possible and allow you opportunities to ask us about assessment marks, but please do not haggle about them.
- Please don't cheat!

**You $\rightarrow$ Us**

- We will strive to be prompt in replying to your requests.
- We provide support and help for you to achieve the best learning outcomes as you can.

# Communication Channels

**Us → You**

- Announcements are made regularly on Canvas.
- Emails will be sent out on a need basis.
- Lecture announcements.

**You → Us**

- Consultation hours, email, before/after class, during labs.
- Official consultation hours: see canvas.

**Many → Many**

- Discussion Board on Canvas. Please don't hesitate to reply to queries on discussion boards.

The programming language used for the course will be **Python**. Deep learning models will be implemented with **PyTorch**.

Infrastructure:

- Google Colab
- AWS Env via RMIT RACE Hub: For Assessments (compatible with Colab).

More details on the above platforms and information to get started is provided in week 1 lab exercises (self-study).
PyTorch tutorial - Week 02 canvas module.

## Academic Integrity

- Please see:
  https://www.rmit.edu.au/students/student-essentials/rights-and-responsibilities/academic-integrity
- You are encouraged to form groups to solve problems; however, *when writing or programming, write in your own words or code, and provide your own solutions.*
- This is not about penalising or yielding the stick, but you've made a conscious act to enroll in this course (thank-you), and we want you to:
  - Learn by trying things, making mistakes
  - Fair for everyone
  - Quality control of your degree